



Surgical management of Crohn disease

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INTRODUCTION

Crohn disease is a chronic, unremitting, incurable inflammatory disorder that can affect any segment of the gastrointestinal tract as well as extraintestinal sites [1]. The mainstay of treatment for Crohn disease is medical. However, one in two patients with Crohn disease can expect to have at least one surgical procedure due to complications or refractory symptoms.

The surgical and endoscopic management of Crohn disease of the small bowel, colon, and rectum is discussed in this topic. The clinical manifestations and diagnosis of Crohn disease, medical therapy for patients with Crohn disease, and extraintestinal manifestations of Crohn disease are discussed elsewhere:

- (See "[Clinical manifestations, diagnosis, and prognosis of Crohn disease in adults](#)".)
- (See "[Overview of the medical management of mild \(low risk\) Crohn disease in adults](#)".)
- (See "[Medical management of moderate to severe Crohn disease in adults](#)".)
- (See "[Perianal Crohn disease](#)".)

PREOPERATIVE PREPARATION

Medical optimization — Although the majority of patients requiring operative intervention for Crohn disease are young, those who present acutely (eg, with sepsis or

perforation) can be seriously ill, and those who present with chronic symptoms (eg, with strictures) can be malnourished.

Therefore, the patient's medical condition should be optimized by correcting anemia, fluid depletion, electrolyte imbalance, and malnutrition prior to surgery if possible. Although some patients may require total parenteral nutrition (TPN) due to prolonged intolerance of oral nutrition, long-term preoperative TPN should be avoided. In patients with fistulizing disease, the use of exclusive enteral nutrition (EEN) has been shown to reduce postoperative septic complications [2]. The use of TPN in surgical patients is discussed separately. (See ["Overview of perioperative nutrition support"](#) and ["Postoperative parenteral nutrition in noncritically ill adults"](#).)

Immunosuppressive or biologic therapy — Most patients with Crohn disease who require surgery are on high-dose glucocorticoid and/or one of the immunomodulators, biologics, or small molecule agents, which can impact surgical outcomes. (See ["Medical management of moderate to severe Crohn disease in adults"](#).)

- Before elective surgery, preoperative high-dose glucocorticoids should be weaned to <20 mg per day before surgical intervention if possible as higher doses can increase the risk of postoperative infectious complications [1]. In patients who require emergency surgery, glucocorticoids need to be tapered gradually after surgery. (See ["The management of the surgical patient taking glucocorticoids"](#) and ["Approach to discontinuing systemic glucocorticoid therapy in adults and children"](#).)
- Most immunomodulators (eg, [6-mercaptopurine](#) or [azathioprine](#)) can be continued until surgery without negative sequelae (eg, postoperative infectious complications) [3].
- The impact of preoperative biologic agents upon surgical outcomes has been controversial [4]. In some studies, preoperative use of [infliximab](#) did not increase postoperative complication rates [5-10]. Some other studies have suggested negative effects of biologic agents such as [vedolizumab](#) [11] but not [ustekinumab](#) [12]. In other studies, patients who received glucocorticoids or infliximab preoperatively had more postoperative complications and unplanned readmissions [13-16]. There is also evidence that higher preoperative serum levels of biologic agents correlate with worse surgical outcomes [14]. The PUCCINI trial showed that the rates of overall infectious complications or surgical site infections did not differ between patients and controls with previous exposure to anti-tumor necrosis factor agents or detectable serum levels of the drugs [17]. Thus, it appears safe to proceed with elective surgery in patients on biologic therapy when otherwise appropriately medically optimized.

- Robust data on the impact of small molecule agents on surgical outcomes remain limited. Compared with previous reviews of overall complications of inflammatory bowel disease surgery, complications after abdominal surgery with preoperative [tofacitinib](#) use do not appear to be negatively affected [18].

Specific complications related to drugs commonly used to treat inflammatory bowel disease are summarized in this table ([table 1](#)) and discussed elsewhere. (See "[Medical management of moderate to severe Crohn disease in adults](#)".)

Abdominal imaging — Abdominal imaging studies are essential in determining the anatomical distribution and complexities of Crohn disease. Both computed tomography enterography (CTE) and magnetic resonance enterography (MRE) are highly accurate in assessing lesions and complications (eg, abscess, fistula) of Crohn disease [19-22]. MRE has the added benefit of not requiring radiation [23]. (See "[Radiation-related risks of imaging](#)", [section on 'Estimating risk from radiation exposure'](#).)

One important distinction that must be made in patients with stricturing Crohn disease is whether the disease is primarily inflammatory, fibrostenotic, or both. An inflammatory stricture or bowel obstruction is usually characterized by surrounding inflammation on imaging studies. A fibrostenotic stricture or obstruction lacks such inflammation but often has bowel dilation upstream of or in between strictured segments (pseudosacculation) [24-26]. Predominantly inflammatory or fibrostenotic strictures are treated differently as discussed below. (See '[Stricturing diseases](#)' below.)

SURGICAL APPROACHES

Common procedures used to treat Crohn disease of the small bowel, colon, and rectum include bowel resection, strictureplasty, and endoscopic balloon dilatation. (See '[Surgical techniques](#)' below.)

The choice of procedures depends upon the indication for operative intervention and the location(s) of the disease (small bowel versus colorectal).

Penetrating diseases — Approximately 4 to 23 percent of Crohn patients present with a penetrating phenotype characterized by perforation, abscess, or fistulization [27].

Perforation — Emergency surgery is required to control sepsis or peritonitis resulting from a bowel perforation. In 1 to 16 percent of patients undergoing surgery for Crohn disease, bowel perforation was the main indication [28].

Patients with a free perforation of the small bowel or colon should undergo immediate surgical resection of the perforated segment. Patients who are hemodynamically unstable or who have edematous bowels, significant intra-abdominal contamination from the perforation, or other risk factors (eg, malnutrition or chronic glucocorticoid use) should have a diverting stoma rather than a primary anastomosis. Stable patients without the above risk factors may undergo primary bowel anastomosis. (See '[Small bowel or ileocecal resection](#)' below and '[Colorectal resection](#)' below.)

Abscess — Patients with an intra-abdominal abscess resulting from Crohn disease should receive antibiotic treatment (see '[Antimicrobial approach to intra-abdominal infections in adults](#)', section on '[Approach to empiric antibiotic selection](#)') and either percutaneous or surgical drainage of the abscess, followed by surgical resection of the involved bowel segment.

- **Percutaneous drainage** – Percutaneous drainage is the preferred treatment for intra-abdominal abscesses [29]. Percutaneous drainage can be performed transgluteally or transabdominally depending upon the location of the abscess. It may be repeated as necessary for any residual or recurrent collections. In contemporary practice, approximately 70 percent of Crohn-related abscesses are managed with percutaneous drainage [30-32].

Successful percutaneous drainage of the abscess resolves the intra-abdominal sepsis, after which the involved bowel segment can be resected in a clean rather than contaminated operation. Percutaneous drainage also permits the patient's nutritional status to be improved prior to surgery and decreases their glucocorticoid use perioperatively.

Following the resolution of an intra-abdominal abscess, the involved bowel segment is typically resected, or else at least 30 percent of patients will develop recurrent abscesses or other symptoms. Patients with long-segment disease, however, may be managed medically [33] to avoid postsurgical complications such as diarrhea or even short bowel syndrome.

The optimal timing of bowel resection should be determined individually for each patient. Most surgeons wait five to seven days after a successful percutaneous drainage before operating to allow local sepsis associated with the abscess to resolve [34].

Abdominal wall abscesses involving the rectus sheath and retroperitoneal abscesses involving the psoas muscle are less common but more difficult to control than intra-

abdominal abscesses. In a retrospective review, all 13 abdominal wall abscesses eventually required surgical drainage after multiple failed attempts of percutaneous drainage [31].

- **Surgical drainage** – A surgical drainage procedure is performed if percutaneous drainage is not feasible or fails to control the sepsis. For those undergoing surgical drainage of an abscess, whether to perform a concomitant bowel resection is determined by operative findings and the patient's condition [29]. Patients who are dependent on glucocorticoids preoperatively should not undergo concomitant bowel resection with abscess drainage, because of the high risk of anastomotic complications. In a retrospective review of 173 patients undergoing ileocecal or ileocolic resection for Crohn disease, preoperative glucocorticoid use (odds ratio [OR] 2.67, 95% CI 1.0-7.2) and intra-abdominal abscess (OR 3.4, 95% CI 1.2-9.8) were two independent predictors of anastomotic complications [35]. Patients with both risk factors developed significantly more anastomotic complications than those who only had an abscess (40 versus 14 percent). As preoperative use of biologic agents (eg, [infliximab](#)) has not been shown to negatively impact postoperative outcomes [17], intraoperative decision making should not be significantly influenced by the presence of biologic agent therapy. (See '[Immunosuppressive or biologic therapy](#)' above.)

Fistula — Transmural bowel inflammation is associated with the development of sinus tracts. Sinus tracts that penetrate the serosa can give rise to fistulas. About 15 percent of patients with Crohn disease develop enteric fistulas [36], which most often connect the intestine to the bladder (enterovesical), skin (enterocutaneous), bowel (enteroenteric), or vagina (enterovaginal). (See "[Medical management of moderate to severe Crohn disease in adults](#)", section on '[Fistulizing disease](#)'.)

Although biologic agents are effective against fistulizing Crohn disease in about half of patients [37,38], refractory fistulas require resection of the involved bowel segment and oversewing of the fistulous opening into the adjacent organ. Resection of the adjacent organ is not necessary unless that organ is also affected by Crohn disease. Intestinal bypass should be avoided because persistent disease in the bypassed segment can lead to complications such as abscess formation, bleeding, perforation, bacterial overgrowth, or malignancy.

Crohn disease also causes perianal fistulas, which are treated differently from enteric fistulas. Perianal fistulas related to Crohn disease are discussed in another topic. (See "[Perianal Crohn disease](#)".)

Strictureing diseases — The transmural inflammatory process that characterizes Crohn disease often results in fibrotic strictures that can lead to bowel obstruction. Although the strictureing phenotype is only seen in 5 to 24 percent of Crohn patients [27], it is most responsible for the development of short gut [39]. Hence, the concept of bowel-preserving surgery is most applicable to diffuse strictureing Crohn disease [40].

Strictures in Crohn patients can be inflammatory, fibrostenotic, or both. It is paramount to distinguish them by imaging (see '[Abdominal imaging](#)' above) since treatments are very different: inflammatory strictures are primarily treated medically (see "[Medical management of moderate to severe Crohn disease in adults](#)"), whereas fibrostenotic strictures are treated with endoscopic dilation or surgically.

Small bowel stricture — Small bowel strictures respond to endoscopic dilatation, strictureplasty, or surgical resection. The best approach is chosen based upon a stricture's length and complexity as well as the patient's intestinal reserve ([algorithm 1](#)):

- Symptomatic, short-segment (<5 cm) small bowel or anastomotic strictures that are not associated with any inflammation or penetrating disease can be treated with surgery or endoscopic dilatation, provided that the location of the stricture is accessible to endoscopic intervention. (See '[Endoscopic dilatation](#)' below.)
- Longer (≥5 cm), multifocal, complicated strictures and strictures that have failed endoscopic therapy require surgical treatment [1]:
 - Bowel resection can be performed to resect a single stricture or multiple strictures within a short segment of the bowel, especially during the index surgery when the bowel length is otherwise normal. Bowel resection is also performed when strictureplasty is contraindicated (eg, perforation, abscess, fistula, malignancy, severe malnutrition). (See '[Small bowel or ileocecal resection](#)' below.)
 - Strictureplasty is preferred in patients with diffuse or recurrent strictures in the duodenum, jejunum, or neoterminal ileum (after ileocecal resection), especially in those with existing or impending short bowel syndrome. Strictureplasty is contraindicated in the presence of an inflammatory mass, thick and unyielding bowel wall, malignancy, active bleeding, or severe malnutrition [1,39,40]. (See '[Small bowel strictureplasty](#)' below.)
 - Heineke-Mikulicz strictureplasty ([figure 1](#)) can be performed for strictures up to 10 cm in length (typically 5 to 6 cm).

- Finney strictureplasty ([figure 2](#)) can be performed for strictures 10 to 20 or 25 cm in length (typically 10 to 12 cm).
- Side-to-side isoperistaltic strictureplasty (SSIS) ([figure 3](#)) can be performed for strictures >20 to 25 cm in length.

Colorectal stricture — Colonic strictures occur in 9 to 13 percent of patients with Crohn disease. They typically occur at a single site and can be associated with dysplasia or cancer in 2 to 6 percent of cases [41]. (See '[Malignancy](#)' below.)

Colorectal strictures that are associated with an anastomosis are typically treated with endoscopic dilatation (see '[Endoscopic dilatation](#)' below). Other Crohn-related colorectal strictures should be assessed endoscopically with biopsy or brushing; strictures that do not permit endoscopic surveillance are surgically resected [42]. Strictureplasty is not usually performed for colorectal strictures [43]. (See '[Colorectal resection](#)' below.)

Malignancy — Longstanding chronic inflammation associated with Crohn disease can cause intestinal mucosa to undergo malignant transformation. Suspicion of a malignant stricture or fistula in patients with Crohn disease should prompt a resection of the bowel segment including the lesion instead of a strictureplasty or intestinal bypass. (See '[Small bowel or ileocecal resection](#)' below and '[Small bowel strictureplasty](#)' below.)

Patients with Crohn disease are more likely to develop malignant or premalignant lesions of the colon or rectum than of the small bowel. Such lesions include cancer, high-grade dysplasia, and multifocal low-grade dysplasia. Patients with longstanding Crohn disease should undergo regular colonoscopic surveillance starting no later than eight years from the onset of symptoms [44].

The current management of Crohn patients with diagnosed colonic malignancy is as follows [1] (see "[Surveillance and management of dysplasia in patients with inflammatory bowel disease](#)"):

- Patients with visible dysplasia may undergo endoscopic resection and continued endoscopic surveillance. However, if the lesion is endoscopically unresectable, multifocal, or found in surrounding flat mucosa, total colectomy or proctocolectomy is indicated.
- Patients with invisible dysplasia on random biopsy should be referred for high-resolution colonoscopy and chromoendoscopy. If repeat biopsy again shows invisible dysplasia, total colectomy or proctocolectomy is indicated.
- Patients diagnosed with adenocarcinoma of the colon should undergo a total colectomy

or proctocolectomy.

The choice between total colectomy and proctocolectomy depends on whether there is rectal involvement or rectal sparing of the disease. (See '[Colorectal resection](#)' below.)

Hemorrhage — Patients with Crohn disease often have guaiac-positive stools, but acute lower gastrointestinal bleed, which is typically from ulcerative colitis, is unusual and occurs in 1 to 10 percent of patients [45]. Hemodynamically stable patients with significant gastrointestinal bleeding from Crohn disease may be managed endoscopically or angiographically.

Unstable patients require urgent bowel resection. If the source of bleeding can be localized by imaging or intraoperative measures, a targeted resection is performed [46]. In cases of persistent hemodynamic instability or serious bleeding that cannot be localized to one segment of the colon, a total

colectomy may be necessary [47]. (See '[Colorectal resection](#)' below.)

Inflammation — Approximately 56 to 81 percent of Crohn patients present with an inflammatory phenotype [27], which is most often treated medically. However, such patients may require operative intervention if they do not respond to, develop complications from, or are noncompliant with medical therapy. (See "[Medical management of moderate to severe Crohn disease in adults](#)".)

Small bowel disease — Patients with medically refractory Crohn enteritis typically require small bowel resection because strictureplasty and endoscopic dilation are contraindicated in the presence of active inflammation. (See '[Small bowel or ileocecal resection](#)' below.)

Colorectal disease — Patients who develop acute fulminant colitis or toxic megacolon due to Crohn disease require urgent surgery to prevent sepsis, perforation, or death. A total colectomy with end ileostomy is the procedure of choice in this life-threatening situation. (See "[Toxic megacolon](#)", section on '[Etiology-specific therapy](#)'.)

The surgical management of Crohn colitis or proctitis with a less acute presentation depends upon the location of the disease:

- For patients with Crohn colitis, both segmental colectomy and total colectomy with ileorectal anastomosis are effective treatments. The choice depends upon the focality of the disease. (See '[Colectomy](#)' below.)

- For patients with Crohn disease confined to the anorectum and the perineum, a complete proctectomy with an intersphincteric approach is the surgical treatment of choice. (See '[Proctectomy](#)' below.)
- For patients with Crohn disease involving both the colon and rectum, a total proctocolectomy with end ileostomy is required. A restorative procedure, such as the ileal pouch-anal anastomosis (IPAA), may be offered [1]. However, performing IPAA for known Crohn disease remains controversial. (See '[Proctocolectomy](#)' below.)

Growth retardation — Crohn disease starts during childhood or adolescence in up to 20 percent of patients, 10 to 40 percent of whom have growth impairment due to chronic intestinal inflammation. Prepubertal patients with Crohn disease require surgery, most commonly bowel resection, if they have growth retardation despite medical therapy [48]. (See "[Growth faltering and pubertal delay in children with inflammatory bowel disease](#)" and '[Small bowel or ileocecal resection](#)' below.)

SURGICAL TECHNIQUES

Common procedures used to treat Crohn disease of the small bowel, colon, and rectum include bowel resection, strictureplasty, and endoscopic balloon dilatation. The indications for each procedure have been described above (see '[Surgical approaches](#)' above); technical details are discussed in this section.

Endoscopic dilatation — Small bowel or anastomotic strictures can be dilated endoscopically with a hydrostatic balloon in select patients who wish to avoid surgery (eg, patients maintained on [infliximab](#) who develop high-grade strictures) or those with a short length of remaining small bowel after prior surgery, provided that the strictures are amenable to endoscopic intervention [49,50].

In two meta-analyses, balloon dilatation was technically successful in 86 to 91 percent of patients and clinically successful (relieved symptoms) in 58 to 70 percent of patients [51,52]. Two to three percent of patients developed perforation after balloon dilatation. At three and five years, 42 and 75 percent of patients required surgery for recurrent diseases.

Balloon dilatation is more likely to achieve long-term success for short strictures (defined as ≤ 4 cm [51] or ≤ 2 cm [53]). Neither active disease at the time of dilatation nor medical therapy afterward predicts the need for repeat dilatation or surgery after balloon dilatation [54].

Patients who require salvage surgery after an unsuccessful balloon dilatation may have more complications compared with those who had surgery as the initial treatment. In a prospective, nonrandomized study of ileocolonic strictures due to recurrent Crohn disease, 114 patients underwent surgery as their initial therapy; 80 patients had salvage surgery after failed endoscopic balloon dilatation [55]. Compared with patients who had initial surgical therapy, those who had salvage surgery developed more surgical site infections (15 versus 5 percent) and required more ileostomies (16 versus 6 percent), despite the mean stricture length being significantly longer in the initial surgery group (4 versus 11 cm).

In small studies, intralesional steroid injection after balloon dilatation has been shown to improve outcomes in children but not adults. In one randomized trial, 29 children with Crohn strictures underwent balloon dilatation with or without intralesional injection of glucocorticoids [56]. Fewer patients who received injections required repeat dilatations (1 in 15 versus 5 in 14 patients) or surgery (0 in 15 versus 4 in 14 patients). However, glucocorticoid injection did not improve outcomes in a pilot trial of adult patients [57].

Small bowel or ileocecal resection — In patients with short-segment stricturing or fistulizing disease, bowel resection is the most efficient way to restore health and improve quality of life [58,59]. A small bowel resection is performed when a segment of small bowel is inflamed or perforated or when there is an abscess or fistula to an adjacent organ. An ileocecal resection is performed when the terminal ileum is involved with Crohn disease.

Margins of resection — Because Crohn disease often recurs, bowel resection should be performed with the goal of preserving as much bowel as possible. Gross inspection, rather than histopathology, should determine resection margins since microscopic disease at the margins has not been associated with an increased risk of recurrence. The presence of limited aphthous ulcers at the margins does not mandate further resection. When necessary, intraoperative enteroscopy can be used to assess the bowel lumen for stricture or ulcerative disease that may be involved in resection margins [60].

In a 1996 trial of 152 patients undergoing ileocecal resection for Crohn disease, recurrence was not affected by the width of the margin of resection from macroscopically involved bowel (25 percent with 2 cm margins versus 18 percent with 12 cm margins) [61]. Recurrence rates also did not increase when microscopic disease was present at the resection margins.

Anastomotic technique — Following a bowel resection, the type of anastomosis (eg, side to side or end to end) performed is determined by surgeon preference as one type

of anastomosis has not been conclusively shown to be superior to another based on available data [62].

For technical reasons, hand-sewn anastomoses are typically in an end-to-end configuration, whereas stapled anastomoses are in side-to-side configuration. These two most popular techniques have been compared in multiple trials and nonrandomized comparative studies with most, but not all, studies favoring the stapled technique because of fewer morbidities. The 2020 European Crohn's and Colitis Organization guidelines recommend a stapled side-to-side small bowel or ileocolonic anastomosis after a bowel resection [63]. However, we feel that the decision should be left to the operating surgeon as there are clinical scenarios in which a stapled anastomosis is not feasible or ill advised. This is also the American Society of Colorectal Surgeons (ASCRS) position [1].

The Kono-S anastomosis is a functional end-to-end anastomosis that is purposefully designed for bowel resection in Crohn patients [64,65]. To perform a Kono-S anastomosis, bowel resection is carried out with GIA staplers placed perpendicular to the mesentery ([figure 4](#)). The bowel stumps are first reinforced with absorbable sutures, then sutured together to create a common support column, a unique feature of this technique. Bowels on both sides of the center support column are then opened longitudinally at the antimesenteric border starting at 1 cm from the support column. The longitudinal incisions are then hand-sewn closed transversely using absorbable sutures in a single-layer manner to complete the anastomosis. In the completed anastomosis, the support column is positioned between the anastomosis and the mesentery. The potential advantage of this anastomotic configuration is the ability to maintain intestinal diameter and thus prevent distortion or stenosis associated with recurrent strictures, which usually start from the mesenteric side of the lumen [64].

In the first trial comparing the Kono-S anastomosis with conventional side-to-side stapled anastomosis in 79 patients with ileocolic Crohn disease, the Kono-S anastomosis resulted in a significant reduction in postoperative endoscopic (22 versus 63 percent at 6 months), clinical (18 versus 30 percent at 24 months), and surgical recurrence rate (0 versus 5 percent at 24 months) [66].

In a systematic review and meta-analysis of nine studies including the trial described above, the Kono-S anastomosis was associated with a lower incidence of endoscopic (25 versus 67 percent) and surgical recurrence (0 to 3.4 percent versus 15 to 24.4 percent) compared with other types of anastomoses. Complications, particularly anastomotic leak rate, were also lower (1.8 versus 9.3 percent) [67]. Despite these provocative data,

further validation by other trials is required before the Kono-S technique can be recommended as the preferred anastomotic technique for those with Crohn disease.

Mesenteric resection — While the Kono-S technique advocates mesenteric preservation as a key to reducing recurrence, the opposite has been postulated by others that a wider mesenteric resection may preclude diseased mesentery from promoting disease recurrence [68]. The 2024 SPICY trial, however, showed that extended mesenteric resection was not superior to conventional resection with regard to endoscopic Crohn disease recurrence [69].

Laparoscopic versus open approach — Laparoscopic bowel resection has been increasingly used to treat patients with Crohn disease [70-72], including those who develop recurrent disease after open surgery [73]. Compared with open surgery, laparoscopic surgery has the short-term benefits of reduced morbidity, expedited recovery, and lower cost as well as long-term benefits of fewer small bowel obstructions [74] and incisional hernias [71,75]. Thus, a laparoscopic approach to bowel resection is preferred when the appropriate expertise is available.

The benefits of laparoscopic Crohn surgery have been demonstrated in several small randomized trials [76-78] and later illustrated by a large retrospective study of 1917 patients who underwent ileocolonic resection for Crohn disease. Compared with patients who had open surgery, those who underwent laparoscopic resection had fewer major (odds ratio [OR] 0.63, 95% CI 0.43-0.91) and minor (OR 0.58, 95% CI 0.41-0.80) complications as well as a shorter hospital stay (by one day) [71].

In a series of 301 laparoscopic resections performed for Crohn disease, one-fifth required conversion to open surgery, mostly commonly due to dense adhesions, pelvic sepsis with fistulizing disease, large inflammatory mass, or thickened mesentery [79].

The recurrence rate of Crohn disease treated with laparoscopic surgery remains high. In a retrospective study of 89 patients who underwent laparoscopic ileocolonic resection, 61 percent developed recurrence after a median of 13 months (range 1.3 months to 8.7 years) [80]. In another study, the recurrence rate after laparoscopic ileocolic resection for Crohn disease was comparable to that after open surgery [81].

Small bowel strictureplasty — Strictureplasty is an alternative to bowel resection for Crohn patients with strictures of the small bowel, especially those who have lost a significant length of small bowel to previous resections [82]. For patients with bowel obstruction secondary to Crohn disease, a strictureplasty can be performed alone or concomitantly with a bowel resection of a different segment [83,84].

The safety and efficacy of strictureplasty were best illustrated by a meta-analysis of observational studies that included 1112 patients with Crohn disease who underwent 3259 strictureplasties [85]. The sites of the strictureplasties were in the small bowel (94 percent), previous anastomosis (4 percent), duodenum (1 percent), or colon (1 percent). Septic complications (leak/fistula/abscess) occurred in 4 percent of patients. At five years, the overall recurrence rate was 28 percent, although only 3 percent were at the same sites as previous strictures. Two patients developed adenocarcinoma at the site of a previous strictureplasty.

Three types of strictureplasty are performed in contemporary practice [39,40]:

- The Heineke-Mikulicz strictureplasty ([figure 1](#)) is performed most often (in about 85 percent of cases requiring a strictureplasty). Each strictured segment is opened longitudinally at the antimesenteric border, and then the enterotomy is closed transversely. Multiple Heineke-Mikulicz strictureplasties can be performed sequentially. The leak rate is about 6 percent, and the recurrence rate is 25 percent at two years, usually away from the original site.
- The Finney strictureplasty ([figure 2](#)) is performed in about 5 to 10 percent of the cases requiring a strictureplasty. The diseased segment is first opened longitudinally at the antimesenteric border. The bowel is then folded back on itself, and the enterotomy is closed in a side-to-side antiperistaltic manner. There is a 10 percent risk of anastomotic bleeding, and the recurrence rate is <25 percent.
- For more extensive strictures or multiple sequential strictures occurring over a long intestinal segment, a side-to-side isoperistaltic strictureplasty (SSIS) should be performed ([figure 3](#)) [86-90]. SSIS is performed in 5 to 10 percent of the cases requiring a strictureplasty. The bowel and mesentery are first divided at the midpoint of the diseased segment to allow the proximal limb to advance to oppose the distal limb. The two limbs are then opened and the enterotomies are closed in a side-to-side, but isoperistaltic, manner.

The efficacy of SSIS was best illustrated in a prospective observational study of 91 patients who underwent SSIS for Crohn-related strictures; 84 patients were followed for at least 5.5 years [91]. Thirty-seven patients developed a recurrence after 55±37 months, of which 24 were at the same site as the strictureplasty. Nine of the recurrent strictures at the same site as the strictureplasty were treated medically; 15 required surgery (13 with re-strictureplasty and 2 requiring excision of the original strictureplasty).

The durability of SSIS was best illustrated in a retrospective series of 60 patients with a

median follow-up of 11 years (range 1 month to 25 years), in which 86 percent of patients kept the original strictureplasty [92]. Although 61 percent developed a symptomatic recurrence, half were remedied with a revision or repeat strictureplasty.

Colorectal resection — Surgical procedures commonly used to treat Crohn colitis or proctitis include segmental colectomy, total colectomy with ileorectal anastomosis, total proctocolectomy with end ileostomy, and proctectomy. The choice of procedure depends upon the location of the disease and the indication for surgery. (See '[Surgical approaches](#)' above.)

The same set of procedures is also used to treat ulcerative colitis and is discussed in detail elsewhere. (See "[Surgical management of ulcerative colitis](#)", section on '[Surgical options](#)' and "[Abdominoperineal resection \(APR\): Open techniques](#)", section on '[Open surgical technique](#)'.)

Colectomy — Patients with Crohn disease may undergo a segmental or a total colectomy, depending upon the extent of the disease. A segmental colectomy is adequate for treating isolated Crohn disease of the colon, such as a colonic stricture that precludes endoscopic surveillance [93]. Patients with two or more involved colonic segments should undergo a total colectomy. In a meta-analysis that compared segmental colectomy with total colectomy for colonic Crohn disease, the two procedures were equally effective, although patients developed earlier recurrences after a segmental colectomy than after a total colectomy [94]. (See "[Overview of colon resection](#)", section on '[Colon resection](#)'.)

Following a total colectomy, an ileorectal anastomosis (IRA) should be performed if the rectum is not involved with Crohn disease [95,96]. Although the probability of clinical recurrence after IRA was 58 and 83 percent at 5 and 10 years, respectively, the probability of rectal preservation at 10 years was as high as 86 percent [97]. About one-half of patients who had an end ileostomy and a defunctionalized rectum after a total colectomy because of rectal diseases required a secondary proctectomy in 6 to 10 years [97]. (See "[Surgical management of ulcerative colitis](#)", section on '[Total abdominal colectomy with ileorectal anastomosis](#)'.)

Proctectomy — For patients with refractory proctitis without colonic involvement, a proctectomy can be performed [97]. The entire rectum should be removed because cancer can develop in a rectal remnant [98].

In Crohn patients, an intersphincteric dissection with primary closure of the perineal wound, rather than a standard abdominal perineal resection, should be performed to

minimize the risk of a nonhealing perineal wound or sexual dysfunction [99,100]. (See ["Abdominoperineal resection \(APR\): Open techniques"](#).)

Crohn disease of the rectum should not be confused with perianal Crohn disease, which includes anal fistula, fissure, or abscess. Perianal Crohn disease is treated medically or by nonresectional procedures [101], which are discussed in another topic. (See ["Perianal Crohn disease", section on 'Management'](#).)

Proctocolectomy — A total proctocolectomy with end ileostomy is performed in patients with longstanding Crohn disease, who either have disease involvement of both the colon and rectum or have malignant/premalignant lesions in either the colon or the rectum. Total proctocolectomy in properly selected patients has been associated with low morbidity, low risk of recurrence, and a long interval to recurrence [102]. (See ["Colorectal disease"](#) above and ["Malignancy"](#) above.)

POSTOPERATIVE MANAGEMENT

Surgery does not cure Crohn disease. In several studies, the endoscopic recurrence rate was as high as 80 percent at one year after surgery; the clinical recurrence rate was 10 to 15 percent per year [103,104]. Thus, patients will require continued medical therapy or at least regular endoscopic surveillance to detect recurrence. This is discussed in another topic. (See ["Management of Crohn disease after surgical resection"](#).)

Patients with longstanding Crohn disease, especially of the colon or rectum, are susceptible to developing neoplasms either before or after operative intervention. Thus, postsurgical endoscopic surveillance is important for early detection of both recurrent diseases and malignancy [105,106]. Those with dysplasia or malignancy require further surgery (see ["Malignancy"](#) above). This is also discussed in another topic. (See ["Surveillance and management of dysplasia in patients with inflammatory bowel disease"](#).)

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See ["Society guideline links: Crohn disease in adults"](#) and ["Society guideline links: Inflammatory bowel disease in children"](#).)

SUMMARY AND RECOMMENDATIONS

- **Indications for medical or antibiotic therapy** – Medical therapy is the mainstay of treatment for most patients with Crohn disease. Operative management is reserved for those who develop complications (eg, perforation, abscess, fistula, hemorrhage, stricture, fistula, or neoplasm) or have persistent symptoms despite best medical management. (See ['Introduction'](#) above and ['Inflammation'](#) above.)

Patients with an intra-abdominal abscess should be treated with antibiotics and undergo percutaneous or surgical drainage of the abscess, followed by surgical resection of the involved bowel segment after the resolution of sepsis. Percutaneous drainage is preferred to surgical drainage whenever possible. (See ['Abscess'](#) above.)

- **Indications for bowel resection** – Patients with Crohn disease who have bowel perforation, abscess, hemorrhage, fistula, neoplasm, or severe inflammation refractory to medical therapy and children who have growth retardation require surgical resection of the bowel segment(s). (See ['Perforation'](#) above and ['Hemorrhage'](#) above and ['Fistula'](#) above and ['Malignancy'](#) above and ['Inflammation'](#) above and ['Growth retardation'](#) above and ['Small bowel or ileocecal resection'](#) above and ['Colorectal resection'](#) above.)

- Small bowel and ileocecal resection for Crohn disease can be performed open or laparoscopically; we recommend a laparoscopic approach whenever appropriate expertise is available (**Grade 1B**). Compared with open surgery, laparoscopic resection is associated with faster recovery, fewer bowel obstructions, and fewer incisional hernias. (See ['Laparoscopic versus open approach'](#) above.)

- For all small bowel or ileocecal resections, we suggest gross inspection, rather than histopathology, to determine resection margins (**Grade 2C**). A wide or histologically negative margin can reduce intestinal reserve without reducing recurrences.

The anastomotic technique (eg, side-to-side, end-to-end, Kono-S) is determined by clinical scenario and surgeon preference. (See ['Margins of resection'](#) above and ['Anastomotic technique'](#) above.)

- Resection techniques used to treat Crohn colitis or proctitis include segmental colectomy, total colectomy with ileorectal anastomosis, total proctocolectomy with end ileostomy, and proctectomy. The choice depends upon the indication for surgery, the location of the disease, and whether the rectum is involved or spared. (See ['Colorectal resection'](#) above.)

- **Indications for endoscopic dilation or strictureplasty** – Patients with a small bowel or anastomotic stricture can be treated with endoscopic dilatation, small bowel

strictureplasty, or small bowel resection. The choice depends upon the stricture's length and complexity as well as the patient's intestinal reserve ([algorithm 1](#)). (See '[Stricturing diseases](#)' above and '[Small bowel or ileocecal resection](#)' above and '[Small bowel strictureplasty](#)' above and '[Endoscopic dilatation](#)' above.)

- Endoscopic balloon dilatation is effective against short (<5 cm) small bowel or anastomotic strictures without inflammatory or penetrating disease. (See '[Endoscopic dilatation](#)' above.)
- Small bowel strictureplasty is an effective treatment for chronic intestinal stricture without active inflammation or complication. It is most commonly used in patients with existing or impending short bowel syndrome. Conventional strictureplasty techniques (eg, Heineke-Mikulicz ([figure 1](#)) or Finney ([figure 2](#))) are used for focal and short strictures; more extensive strictures or multiple sequential strictures occurring over a long intestinal segment require a side-to-side isoperistaltic strictureplasty ([figure 3](#)). (See '[Small bowel strictureplasty](#)' above.)
- Anastomotic colorectal strictures can be treated with endoscopic dilatation. Other Crohn-related strictures of the colon or rectum are best treated with surgical resection. Strictureplasty of the colon is rarely performed. (See '[Colorectal stricture](#)' above and '[Colorectal resection](#)' above.)

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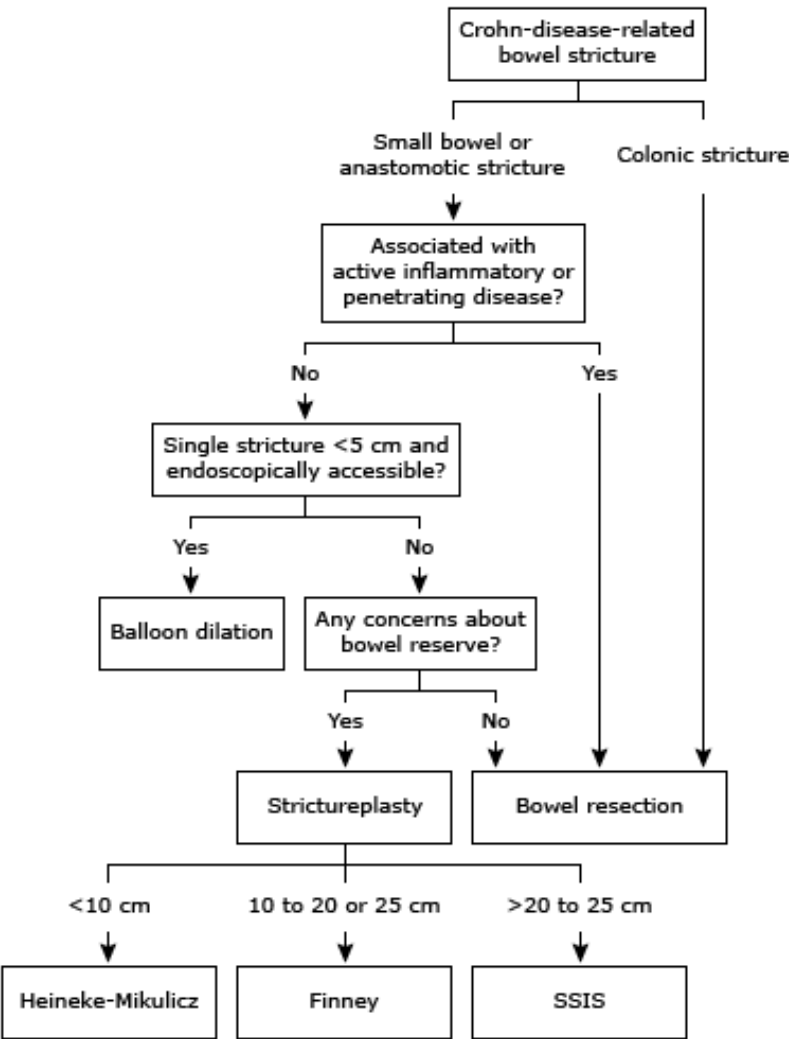
GRAPHICS

Table 1: Complications of medical therapy for IBD

Medication	Perioperative complications
Prednisone	■ Addisonian crisis ■ Depression ■ Adrenal insufficiency
Sulfasalazine	■ Neutropenia ■ Blood dyscrasias ■ Hypersensitivity
Azathioprine and 6-mercaptopurine	■ Leukopenia ■ Thrombocytopenia ■ Macrocytic anemia ■ Bone marrow suppression ■ Hepatotoxicity
Mesalamine	■ Diarrhea ■ Nausea/vomiting ■ Headache
Metronidazole	■ Convulsive disorders ■ Peripheral neuropathy
Cyclosporine	■ Nephrotoxicity ■ Hepatotoxicity
Biologic agents	■ Injection site reactions ■ Infusion reactions ■ Infections ■ Demyelinating disease ■ Heart failure ■ Malignancy ■ Induction of autoimmunity
Small molecules	■ Wound infection ■ Delayed wound healing ■ Thromboembolic events

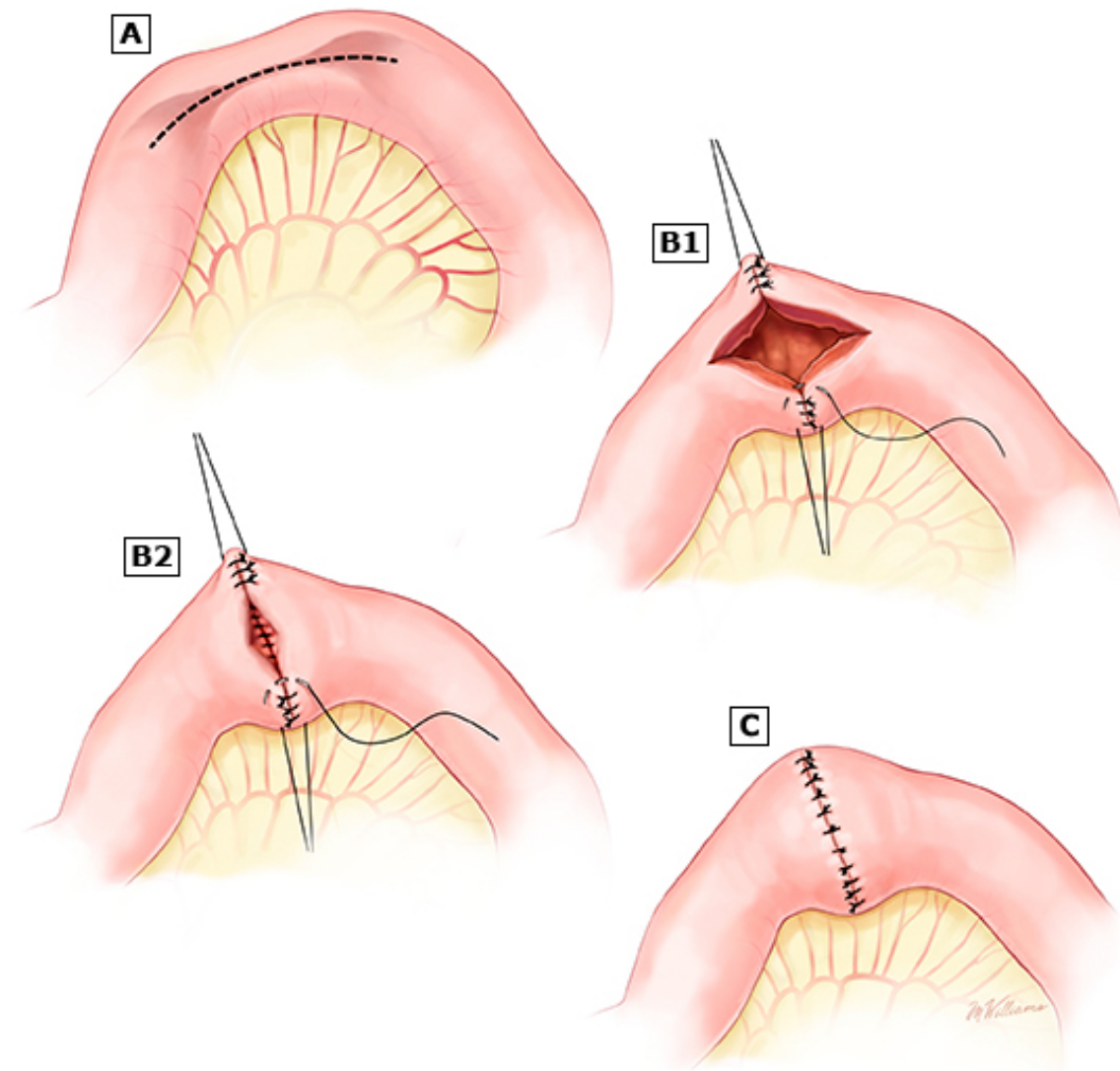
IBD: inflammatory bowel disease.

Algorithm 1: Crohn-related bowel stricture



SSIS: side-to-side isoperistaltic strictureplasty.

Figure 1: Heineke-Mikulicz strictureplasty



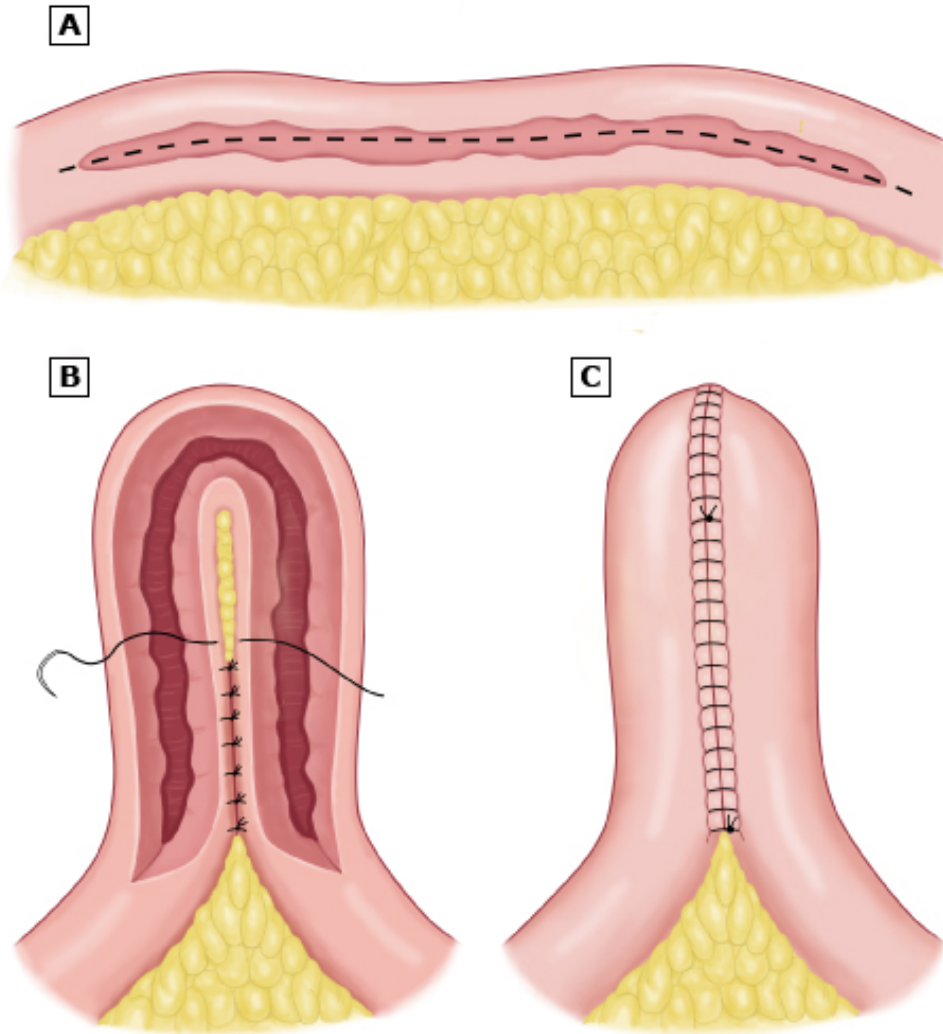
(A) A full-thickness longitudinal incision is made along the antimesenteric border of the stricture for approximately 1 to 2 cm beyond the stricture.

(B1) The lumen is closed in one layer transversely.

(B2) If the surgeon prefers, the lumen can be closed in two layers.

(C) The resulting diameter of the bowel is sufficient to relieve the obstruction and preserve the length of bowel.

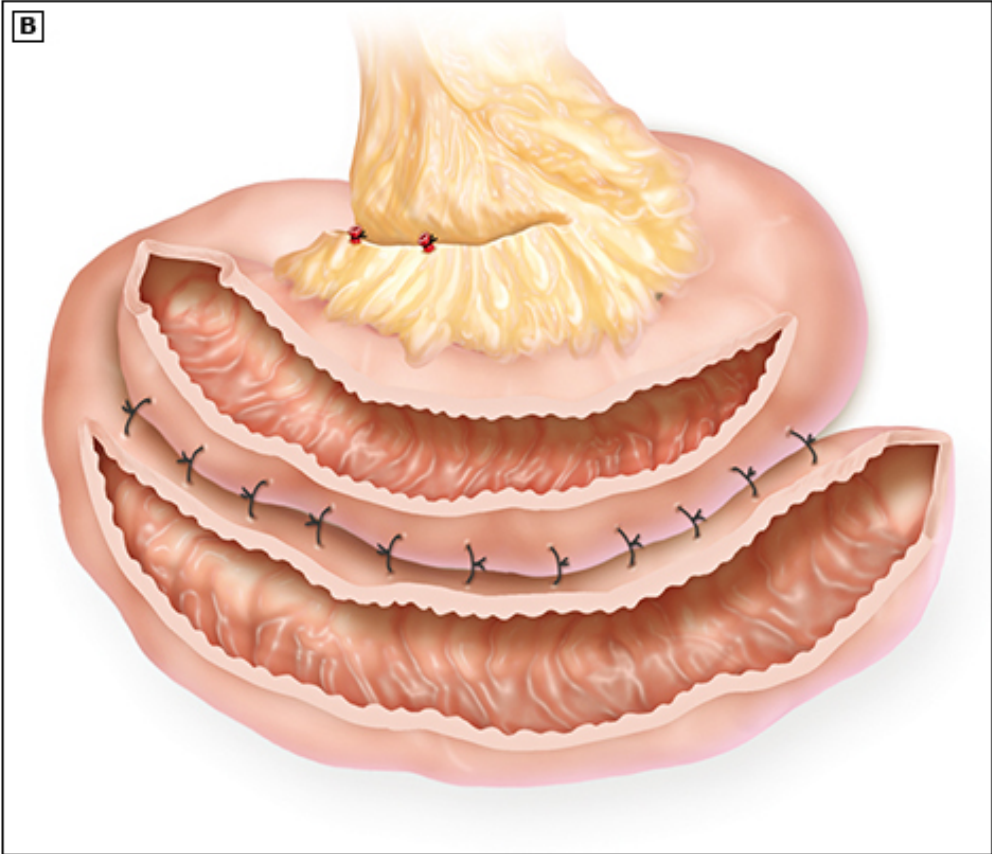
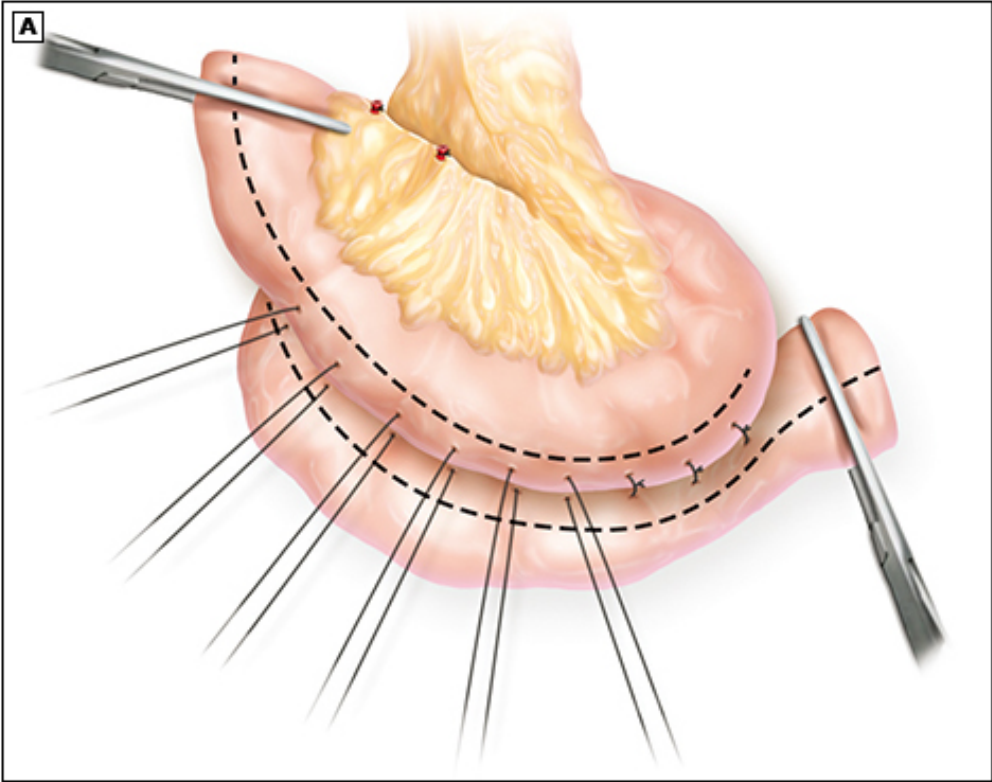
Figure 2: Finney strictureplasty

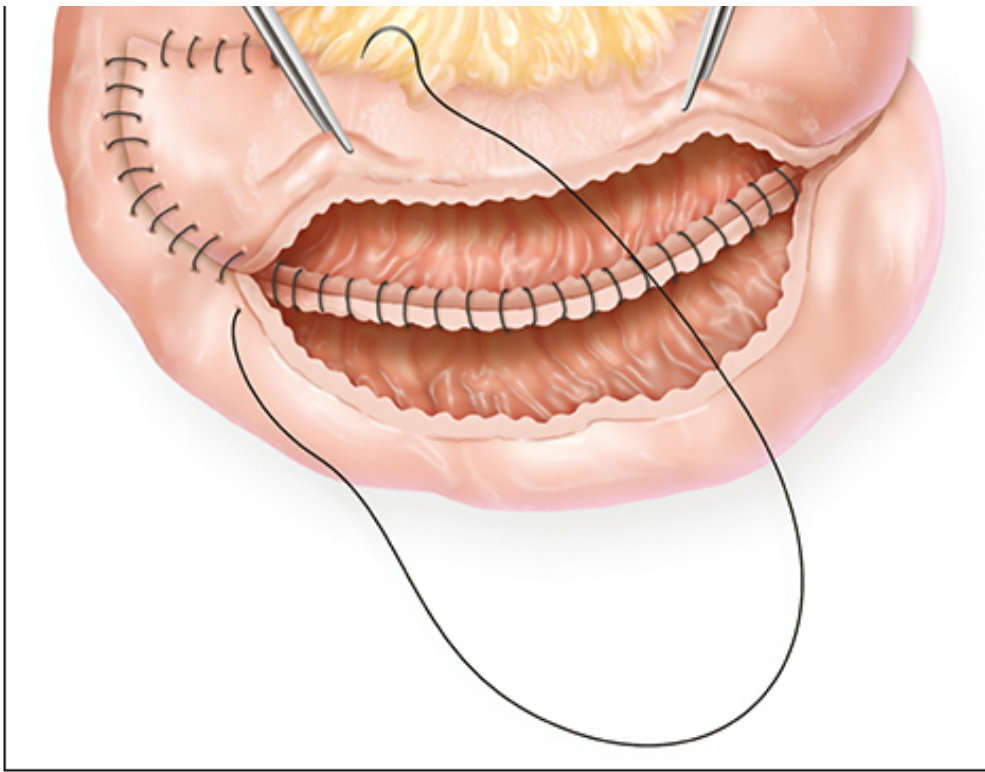


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Figure 3: Side-to-side isoperistaltic strictureplasty





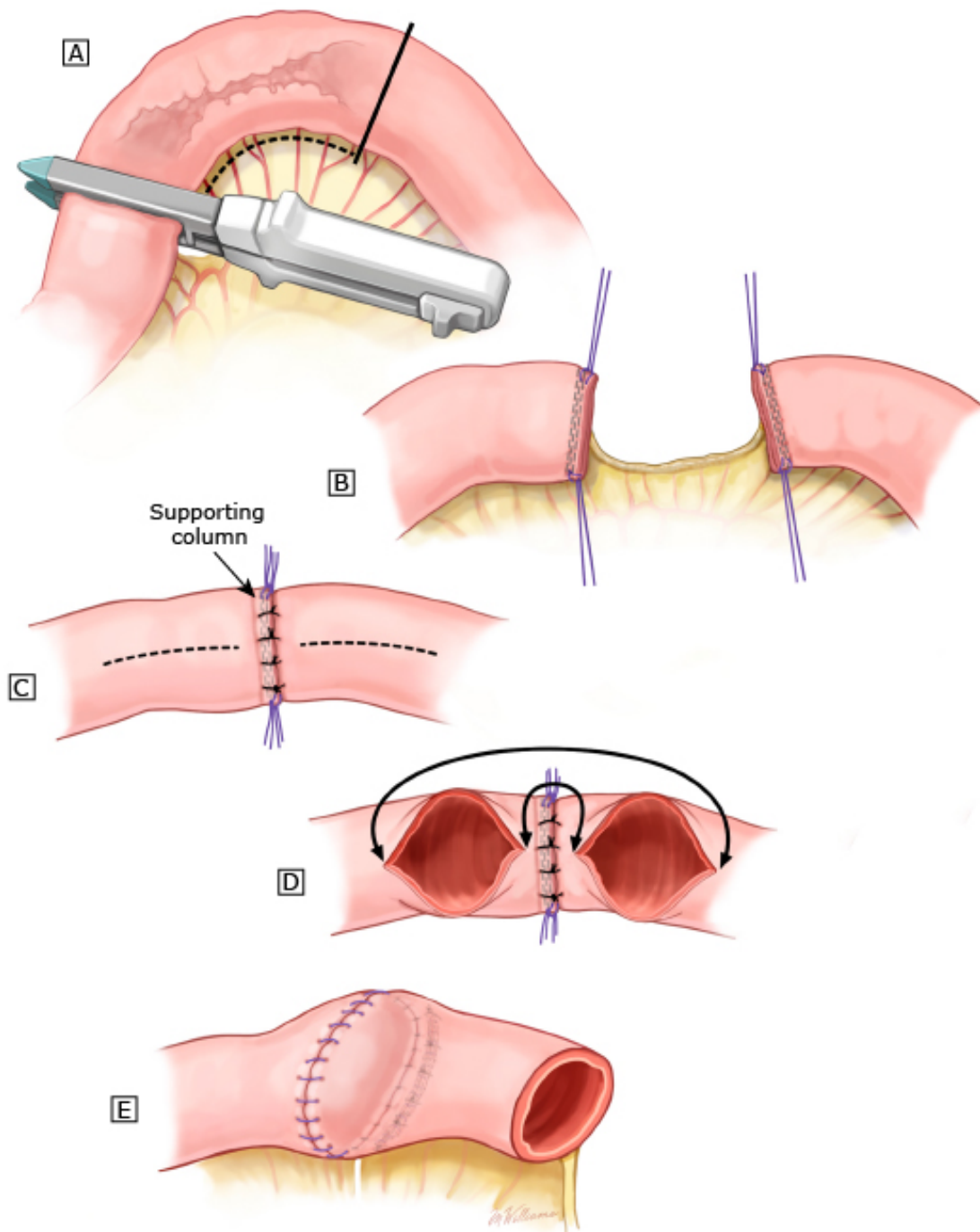
(Panel A) The mesentery of the intestinal loop is divided at its midpoint, and the small bowel is severed between atraumatic intestinal clamps. The proximal intestinal loop is moved over the distal one in a side-to-side fashion such that stenotic areas of one loop are placed adjacent to dilated areas of the other loop. The two loops are approximated by a layer of interrupted nonabsorbable sutures.

(Panel B) A longitudinal enterotomy is performed on both loops, and the intestinal ends are spatulated (tapered) to avoid blind stumps.

(Panel C) The enterotomies are closed with an internal (posterior) running, full-thickness absorbable suture that is continued anteriorly and finished as a running Connell suture. This closure is reinforced by another layer of interrupted seromuscular nonabsorbable sutures (not shown).

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Figure 4: Kono-S anastomosis



To perform a Kono-S anastomosis, bowel resection is carried out with GIA staplers placed perpendicular to the mesentery (A). The bowel stumps are first reinforced with absorbable sutures, then sutured together to create a common support column, a unique feature of this technique (B). Bowels on both sides of the center support column are then opened longitudinally at the antimesenteric border, starting at 1 cm from the support column (C). The longitudinal incisions are then hand-sewn closed transversely using absorbable sutures in a single-layer manner to complete the anastomosis (D). In the completed anastomosis, the support column is positioned between the anastomosis and the mesentery (E). The potential advantage of the support column is to maintain the diameter and dimension of the anastomosis and prevent distortion or stenosis associated with recurrent strictures, which usually starts from the mesenteric side.

Modified from: Kono T, Ashida T, Ebisawa Y, et al. A new antimesenteric functional end-to-end handsewn anastomosis: surgical prevention of anastomotic recurrence in Crohn's disease. *Dis Colon Rectum* 2011; 54:586.

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Contributor Disclosures

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